No.	Solution	Remarks
1(a)	Horizontal component of initial velocity $u_x = 64\cos 7.0^\circ$ Time $t = \frac{s_x}{u_x}$ $= \frac{11.9}{64\cos 7.0^\circ}$ $= 0.1873$ ≈ 0.187 s	[1] for correct substitution [1] for correct answer
1(b)	Method 1: Using $s_y = u_y t + \frac{1}{2}a_y t^2$, Decrease in height from P, $+\downarrow \quad s_y = u_y t + \frac{1}{2}a_y t^2$ $= \; 64\sin 7.0^\circ \quad 0.187 \; + \frac{1}{2} \; 9.81 \quad 0.187 \; ^2$ $= 1.63$ m Final height $= 2.8 - 1.63 = 1.17$ m which is higher than height of the net (1.0 m), so the tennis ball passes over the net. Method 2: Vertical distance to fall to the net $= 2.8 - 1.0 = 1.8$ m Using $s_y = u_y t + \frac{1}{2}a_y t^2$, $+\downarrow \quad 1.8 = \; 64\sin 7.0^\circ \; t + \frac{1}{2} \; 9.81 \; t^2$ $4.905t^2 + 7.8t - 1.8 = 0$ Time to fall this distance $t = 0.204$ s or -1.795 s (reject) which is greater than 0.187 s, so the tennis ball reaches the net before it has fallen far enough. Hence the ball passes over the net.	[1] for correct substitution [1] for decrease in height [1] for final height of ball at net

No.	Solution	Remarks
1(c)	Using $s_y = u_y t + \frac{1}{2} a_y t^2$, $+\downarrow \quad 2.8 = \; 64\sin 7.0^\circ \; t + \frac{1}{2} \; 9.81 \; t^2$ $4.905t^2 + 7.8t - 2.8 = 0$ Time to fall to the ground $t = 0.302$ s or -1.892 s (reject) Horizontal distance from base of net $= \; 64\cos 7.0^\circ \quad 0.302 \; - 11.9$ $= 7.284$ ≈ 7.3 m	[1] for correct substitution [1] for correct time to fall to ground [1] for correct answer